PRIVILEGES

➢ **Grant system privs(Create session, create tablespace) to user SHAUSR.**

STEP 1: Create a user.

```
SQL> CREATE USER SHAUSR IDENTIFIED BY SHAUSER DEFAULT TABLESPACE USERS QUOTA UNLIMITED ON USERS;

User created.

SQL> SELECT USERNAME,DEFAULT_TABLESPACE FROM DBA_USERS WHERE USERNAME='SHAUSR';

USERNAME   DEFAULT_TABLESPACE
---------- ------------------------------
SHAUSR     USERS
```

STEP 2: Grant to the user

```
SQL>  GRANT CREATE SESSION, CREATE TABLESPACE TO SHAUSR;

Grant succeeded.
```

STEP 3: Check the system privilege of the user.

```
SQL> SELECT * FROM DBA_SYS_PRIVS WHERE GRANTEE='SHAUSR';

GRANTEE                        PRIVILEGE                                          ADM COM INH
------------------------------ -------------------------------------------------- --- --- ---
SHAUSR                         CREATE TABLESPACE                                  NO  NO  NO
SHAUSR                         CREATE SESSION                                     NO  NO  NO
```

➢ **Grant system privs(Create session, create tablespace) to user SHAUSR1 with admin option.**

STEP 1: First give the system privilege with admin option.

```
SQL>
SQL> GRANT CREATE SESSION, CREATE TABLESPACE TO SHAUSR WITH ADMIN OPTION;

Grant succeeded.
```

STEP 2: Check the admin option

```
SQL>
SQL> SELECT * FROM DBA_SYS_PRIVS WHERE GRANTEE='SHAUSR';

GRANTEE                        PRIVILEGE                                          ADM COM INH
------------------------------ -------------------------------------------------- --- --- ---
SHAUSR                         CREATE TABLESPACE                                  YES NO  NO
SHAUSR                         CREATE SESSION                                     YES NO  NO
```

➢ **How to check which user has admin privs?**

```
SQL> SELECT * FROM DBA_SYS_PRIVS WHERE ADMIN_OPTION='YES';

GRANTEE                        PRIVILEGE                                          ADMIN_OPTION        COM INH
------------------------------ -------------------------------------------------- ------------------- --- ---
SHAUSR                         CREATE SESSION                                     YES                 NO  NO
SHAUSR                         CREATE TABLESPACE                                  YES                 NO  NO
```

- **Grant object privs(Select , Insert) to user SHAUSR to access and insert HR.Employees tables.**

STEP 1: Grant the object privilege to 'SHAUSR' to access the hr.employees.

```
SQL>
SQL>
SQL> GRANT SELECT, INSERT ON HR.EMPLOYEES TO SHAUSR;

Grant succeeded.
```

STEP 2: Check the object privilege.

```
SQL>
SQL> SELECT GRANTEE,OWNER,TABLE_NAME,GRANTOR,PRIVILEGE FROM DBA_TAB_PRIVS WHERE GRANTEE='SHAUSR';

GRANTEE             OWNER                TABLE_NAME         GRANTOR             PRIVILEGE
------------------- -------------------- ------------------ ------------------- -------------------------
SHAUSR              HR                   EMPLOYEES          HR                  INSERT
SHAUSR              HR                   EMPLOYEES          HR                  SELECT
```

- **Grant object privs(Select, Update) to user SHAUSR1 to access and append HR.Employees table s with Grant option.**

STEP 1: Create a user 'SHAUSR1'

```
SQL> CREATE USER SHAUSR1 IDENTIFIED BY SHAUSR1 DEFAULT TABLESPACE USERS QUOTA UNLIMITED ON USERS;

User created.

SQL> SELECT USERNAME,ACCOUNT_STATUS FROM DBA_USERS WHERE USERNAME='SHAUSR1';

USERNAME   ACCOUNT_STATUS
---------- --------------------------------
SHAUSR1    OPEN
```

STEP 2: Grant the object level permission to shausr1 and append hr.employees**.**

```
SQL> SELECT GRANTEE,OWNER,TABLE_NAME,GRANTOR,PRIVILEGE,GRANTABLE FROM DBA_TAB_PRIVS WHERE GRANTEE='SHAUSR1';

GRANTEE              OWNER                TABLE_NAME          GRANTOR      PRIVILEGE                 GRANTABLE
-------------------- -------------------- ------------------- ------------ ------------------------- ------------
SHAUSR1              HR                   EMPLOYEES           HR           SELECT                    YES
SHAUSR1              HR                   EMPLOYEES           HR           UPDATE                    YES
```

- **Revoke system privs(Create session, create tablespace) from user SHAUSR.**

```
SQL> REVOKE CREATE SESSION, CREATE TABLESPACE FROM SHAUSR;

Revoke succeeded.

SQL> SELECT * FROM DBA_SYS_PRIVS WHERE GRANTEE='SHAUSR';

no rows selected
```

- **Revoke object privs(Select , Insert) from user SHAUSR1 to restrict access HR.Employees tables.**

STEP 1: Check the object privilege to SHAUSR1.

```
SQL> SELECT GRANTEE,OWNER,TABLE_NAME,GRANTOR,PRIVILEGE,GRANTABLE FROM DBA_TAB_PRIVS WHERE GRANTEE='SHAUSR1';

GRANTEE              OWNER                TABLE_NAME         GRANTOR      PRIVILEGE                 GRANTABLE
-------------------- -------------------- ------------------ ------------ ------------------------- ------------
SHAUSR1              HR                   EMPLOYEES          HR           SELECT                    YES
SHAUSR1              HR                   EMPLOYEES          HR           UPDATE                    YES
```

STEP 2: Revoke the SHAUSR1 permission.

```
SQL>  REVOKE SELECT,UPDATE ON HR.EMPLOYEES FROM SHAUSR1;

Revoke succeeded.

SQL> SELECT GRANTEE,OWNER,TABLE_NAME,GRANTOR,PRIVILEGE,GRANTABLE FROM DBA_TAB_PRIVS WHERE GRANTEE='SHAUSR1';

no rows selected

SQL> |
```

- **Create a read user "SHA_READ" and grant Select privs to access all HR Schemas database obje ct (With Script)**

STEP 1: Create a script

```
[oracle@localhost u01]$
[oracle@localhost u01]$ cat user_creation.sql
--------------------------------------------------
-- Step 1: Create Read-Only User
--------------------------------------------------

CREATE USER SHA_READ
IDENTIFIED BY SHA_READ
DEFAULT TABLESPACE USERS
TEMPORARY TABLESPACE TEMP
QUOTA UNLIMITED ON USERS;

--------------------------------------------------
-- Step 2: Grant Login Privilege
--------------------------------------------------

GRANT CREATE SESSION TO SHA_READ;

------------------------------------------------------
-- Step 3: Grant object privillege
-------------------------------------------------------
BEGIN
   FOR r IN (
      SELECT table_name
      FROM dba_tables
      WHERE owner = 'HR'
   )
   LOOP
      EXECUTE IMMEDIATE
         'GRANT SELECT ON HR.'||r.table_name||' TO SHA_READ';
   END LOOP;
END;
/
------------------------------------------------------------------
```

STEP 2: Run the script

```
SQL>
SQL> @/u01/user_creation.sql

User created.


Grant succeeded.


PL/SQL procedure successfully completed.

SQL> select username from dba_users where username='SHA_READ';

USERNAME
--------------------------------------------------------------------------------
SHA_READ
```

STEP 3: Check the object level privilege.

```
SQL> SELECT GRANTEE,OWNER,TABLE_NAME,GRANTOR,PRIVILEGE,GRANTABLE FROM DBA_TAB_PRIVS WHERE GRANTEE='SHA_READ';

GRANTEE    OWNER      TABLE_NAME   GRANTOR      PRIVILEGE                                GRA
---------- ---------- ------------ ------------ ---------------------------------------- ---
SHA_READ   HR         REGIONS      HR           SELECT                                   NO
SHA_READ   HR         COUNTRIES    HR           SELECT                                   NO
SHA_READ   HR         LOCATIONS    HR           SELECT                                   NO
SHA_READ   HR         DEPARTMENTS  HR           SELECT                                   NO
SHA_READ   HR         JOBS         HR           SELECT                                   NO
SHA_READ   HR         EMPLOYEES    HR           SELECT                                   NO
SHA_READ   HR         JOB_HISTORY  HR           SELECT                                   NO

7 rows selected.
```